

UMC 0.11um RFCMOS Al Advanced Enhancement 3.3V MIS Varactor SPICE Model Document

**Version 0.2 Phase 1
2010/07/08**

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1 Introduction

1.1 Revision History

Table 1-1 Revision history

Version	Date	Author	Remark
0.1_p1	2009/03/11	Jian Zhi	Original release.
0.1_p2	2010/04/12	Nancy Lee, Shih-Hsin Yeh	Revise W, L definition to meet the shrink methodology.
0.2_p1	2010/07/08	Laney Fan	1.Modify RF model based on new silicon data. 2.Baseband model was copied from L110AE_NCAP33_V012.

1.2 Model Usage

The compact model is provided in various formats. It is guaranteed that the simulation results of this model from different simulators show differences less than 0.5% and 1% for DC and AC simulations respectively. It should be noted that the model has been verified on different simulators of the specific versions listed below and the simulation results from other versions of the simulators might be different.

Synopsys HSPICE release 2007.9
Cadence Spectre version 6.2.1.299
Mentor Graphics Eldo version 6.11_1.1

To cope with the impact of process variations on circuit performance, worst-case models are provided along with the typical case and they are listed below.

- min: minimum capacitance and quality factor
- typ: typical capacitance and quality factor
- max: maximum capacitance and quality factor

Input Parameters:

- lf: gate finger length in meter
- wf: gate finger width in meter
- m: multiplier
- nf: number of gate fingers in an array

Follow these steps to invoke the model file during simulation.

For Synopsys Hspice

- Copy ModelFileName.mdl and ModelFileName.lib into your simulation directory.
- Add the following statement in the input file to the simulator for 90% dimension shrinkage
.lib "/ModelFileName.lib" CaseName
, where CaseName could be typ, max, or min.
- Define device condition by 'Xxx' statement.

Example:

```
x1 gate nwell psub DeviceName lf=5e-7 wf=5e-6 nf=4 m=1
```

For Cadence Spectre

- Copy ModelFileName.mdl.scs and ModelFileName.lib.scs into your simulation directory.

- Add the following statement in the input file to the simulator for 90% dimension shrinkage

include "./ModelFileName.lib.scs" section=CaseName

, where CaseName could be typ, max, or min.

- Define device condition by 'Xxx' statement.

Example:

x1 (gate nwell psub) DeviceName lf=5e-7 wf=5e-6 nf=4 m=1

For Mentor Graphics Eldo

- Copy ModelFileName.mdl.eldo and ModelFileName.lib.eldo into your simulation directory.

- Add the following statement in the input file to the simulator for 90% dimension shrinkage

.lib "./ModelFileName.lib.eldo" CaseName

, where CaseName could be typ, max, or min.

- Define device condition by 'Xxx' statement.

Example:

x1 gate nwell psub DeviceName lf=5e-7 wf=5e-6 nf=4 m=1

ModelFileName: L110AE_VARMIS33_V021_RF.lib

DeviceName: varmis_33_rf

Geometry description (before 90% shrinkage):

$0.35\ \mu\text{m} \leq \text{lf} \leq 2\ \mu\text{m}$

$2.5\ \mu\text{m} \leq \text{wf} \leq 10\ \mu\text{m}$

$2 \leq \text{nf} \leq 8$ (even number only)

Bias Range:

$-3.3\text{V} \leq \text{VGS} \leq 3.3\text{V}$

$\text{VDS} = \text{VBS} = 0\text{V}$

Temperature range:

$-50\ ^\circ\text{C} \sim +125\ ^\circ\text{C}$

Frequency:

$0.1\text{GHz} \sim 20.1\ \text{GHz}$

2 3.3V N+/Nwell MIS Varactor RF Sub-circuit Model

In this report, RF modeling results of 3.3V N+/Nwell Standard Performance MIS varactors are described.

2.1 Device Architecture and Layout

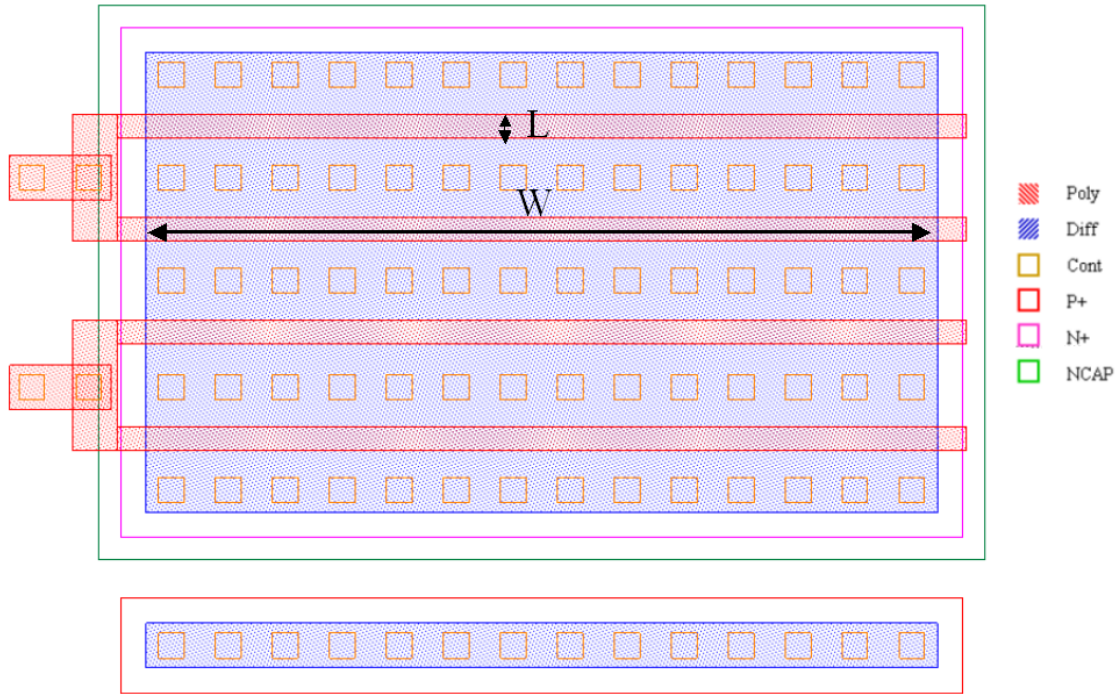


Figure 2-1 Example layout of N+/Nwell MOS varactors

2.2 Sub-Circuit Model Topology

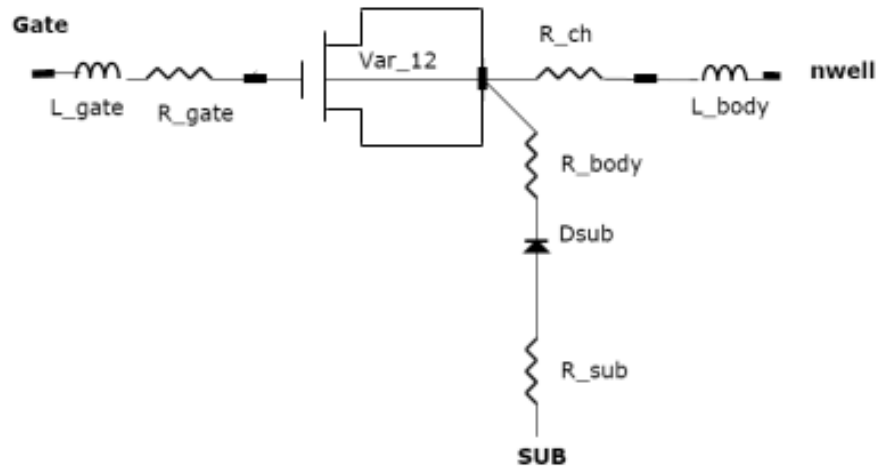


Figure 2-2 Sub-circuit model of N+/Nwell MIS Varactor

2.3 Silicon Wafer for Modeling

Table 2-1 Test wafer information for VARMIS

	Wafer for VARMIS measurements data
Mask ID	BSW0FA
Lot ID	D4YAA
Wafer Number	#2

2.4 Test devices

The following geometries were available for parameters extraction.

Table 2-2 Geometry of N+/Nwell VARMIS devices measured for parameters extraction.

W _{FL}	1.812	0.912	0.675	0.462	0.327
9			X		
4.5	X	X	X	X	X
2.25			X		

Note : The sizes are post-shrinkage.

2.5 Process Corner models

Table 2-3 Corner parameters

Parameter	Min	Max
tox0	Nominal +1.50000e-10	Nominal – 1.50000e-10
toxc	Nominal +1.50000e-10	Nominal – 1.50000e-10
dcw	Nominal – 2.9050e-07	Nominal + 2.8650e-07
dcl	Nominal – 3.0700e-08	Nominal + 2.7050e-08
cmina	Nominal +8.0000e-02	Nominal –6.7200e-02
r_gate	Nominal × 1.15	Nominal × 0.85
cj	Nominal × 1.1	Nominal × 0.9
cjsw	Nominal × 1.1	Nominal × 0.9

2.6 MIS varactor RF model

2.6.1 Model Accuracy Verification

In all measurements (denoted by .m), Port-1 and Port-2 of VNA were connected to gate and nwell, respectively.

**Table 2-4 Comparison of measurements and as-extracted simulation results @
Freq.= 5.9GHz & T=25 °C**

Device	Dimensions				Cgb/Vgb=3.3V		Cgb/Vgb=-3.3V		Tuning ratio		Q/Vgb=0V	
name	Wf	Lf	Nf	m	meas.	simu.	meas.	simu.	meas.	simu.	meas.	simu.
MISVAR_301	4.5	0.327	4	4	1.30E-13	1.29E-13	5.47E-14	5.35E-14	2.37	2.41	22.43	21.86
MISVAR_302	4.5	0.462	4	4	1.75E-13	1.73E-13	6.18E-14	6.60E-14	2.83	2.63	18.90	18.97
MISVAR_303	4.5	0.687	4	2	1.25E-13	1.24E-13	4.42E-14	4.32E-14	2.84	2.87	15.44	15.52
MISVAR_304	4.5	0.912	4	2	1.64E-13	1.61E-13	5.46E-14	5.32E-14	3.00	3.02	12.56	13.34
MISVAR_305	4.5	1.812	2	2	1.57E-13	1.54E-13	4.56E-14	4.62E-14	3.44	3.33	8.99	9.17
MISVAR_306	2.25	0.462	4	8	1.81E-13	1.95E-13	7.13E-14	7.23E-14	2.54	2.70	18.40	16.74
MISVAR_307	9	0.462	4	2	1.77E-13	1.72E-13	6.73E-14	6.55E-14	2.63	2.62	8.71	8.10
MISVAR_308	4.5	0.462	2	8	1.79E-13	1.73E-13	7.21E-14	7.35E-14	2.48	2.36	14.27	13.70
MISVAR_309	4.5	0.462	8	2	1.78E-13	1.73E-13	6.83E-14	6.60E-14	2.61	2.63	14.45	12.40

Note:

- C & Q are extracted from Y12
- $Q = -\text{imag}(Y12^{-1})/\text{real}(Y12^{-1})$
- $C = -(2*\pi*\text{freq}*\text{imag}(Y12^{-1}))^{-1}$
- Wf and Lf in the table 2-4 are post-shrinkage.

2.6.2 Equation to Estimate Maximum Capacitance

The following equation can be used to estimate the capacitance value at (Vgb= 3.3 V) at low frequency (<1GHz) for given dimension (Lf, Wf, Nf and M)

$$C = -0.00362 \times (Wf \times Lf) + 6e-21 * M / Wf + 1.65e-13$$

The unit for capacitance is Farad, for l & w is meter. W & L are post-shrinkage sizes. For example, for a MIS varactor with Lf=0.5um, Wf=5um, Nf=4, M=4, the capacitance value at Vgb=3.3V is about 1.61e-13 (F)

2.6.3 Model verification plots

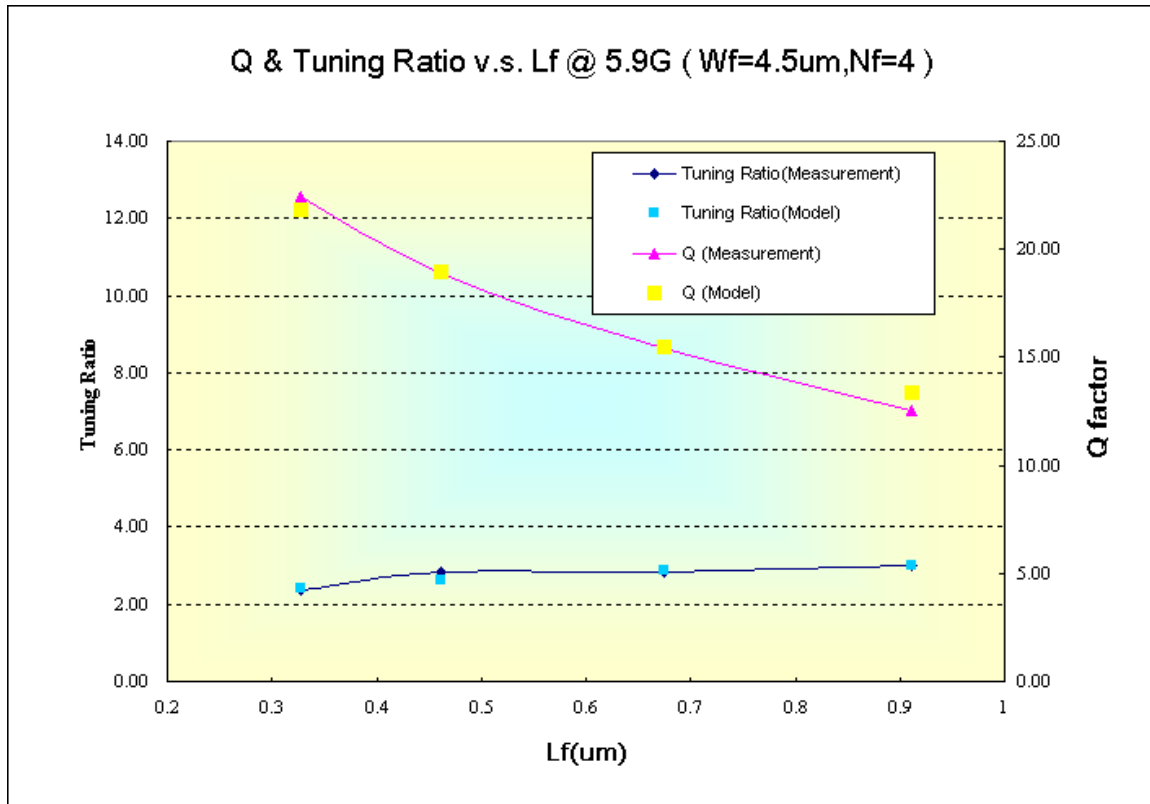


Figure 2-3 Q & Tuning Ratio vs. Lf @ 5.9 GHz

- wf & lf are post-shrinkage sizes.

- Smaller gate length can achieve better Q. Larger gate length can achieve high tuning ratio.

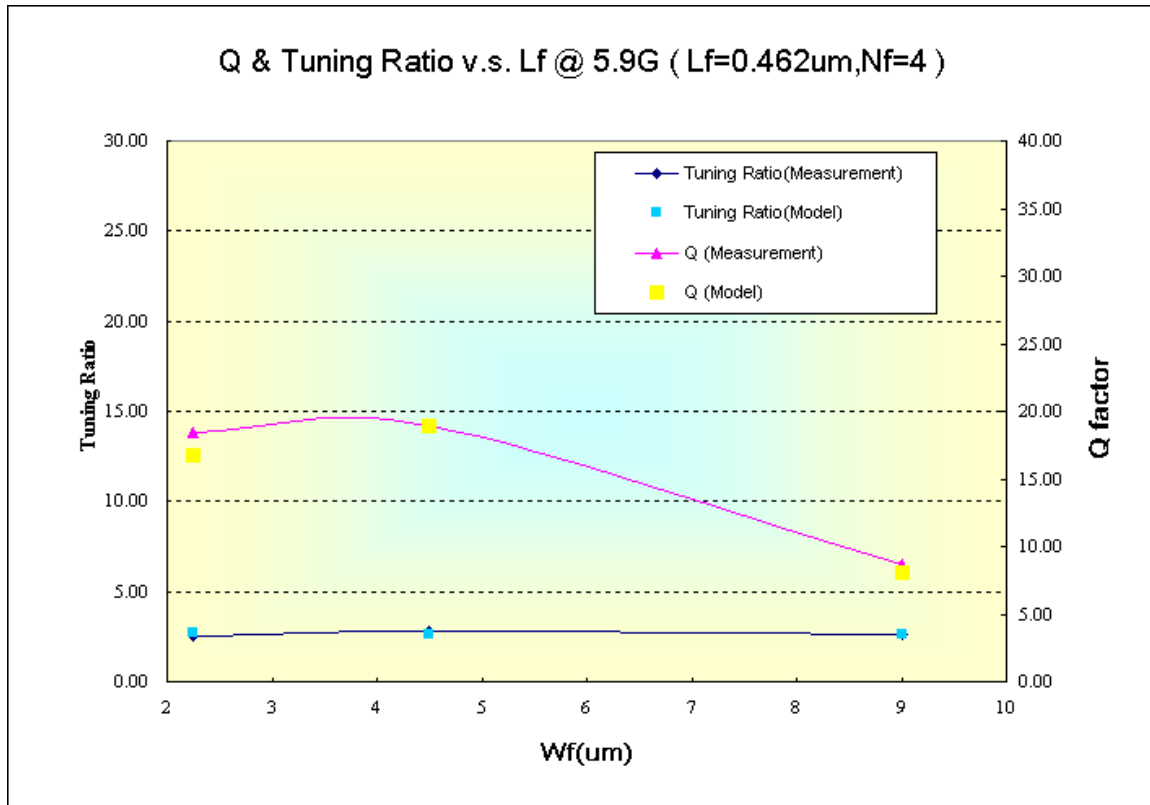


Figure 2-4 Q & Tuning Ratio vs. Wf @ 5.9 GHz

-wf & lf are post-shrinkage sizes.

- Larger gate width can achieve high tuning ratio; smaller gate width can achieve better Q due to reduced gate resistance.

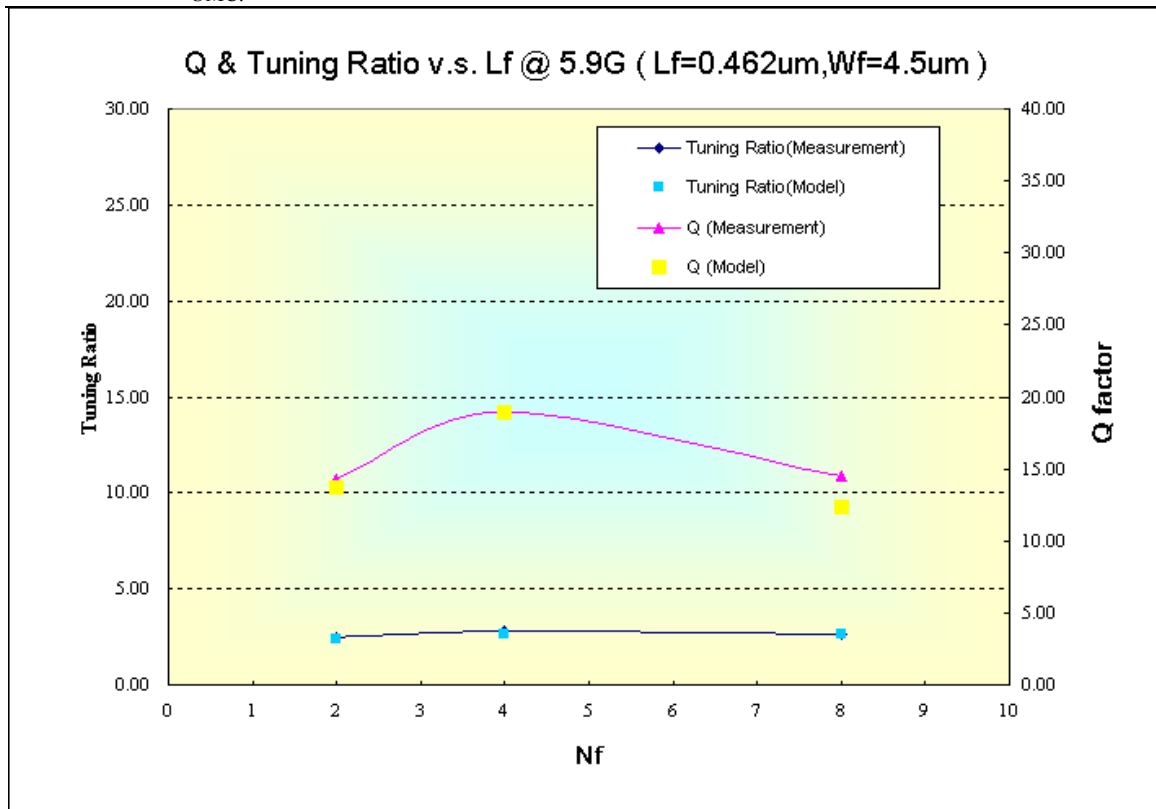


Figure 2-5 Q & Tuning Ratio vs. Nf @ 5.9 GHz

- Tuning ratio is relatively insensitive to finger number.

VAR_301

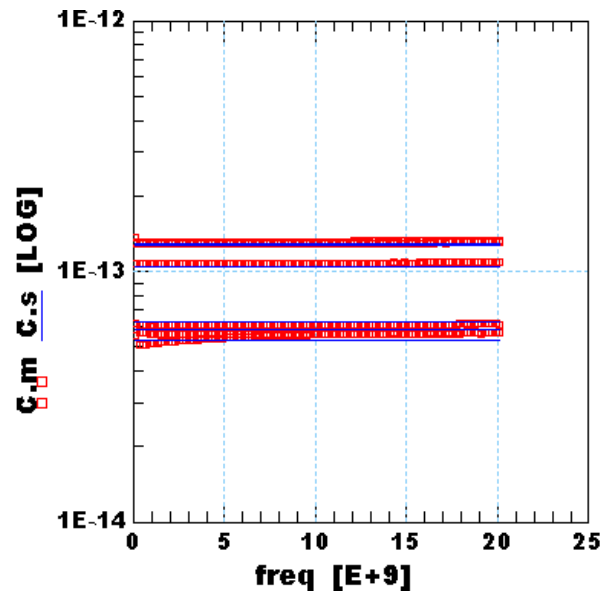


Figure 2-6 VAR_301:Cg vs. Freq at $V_g = -3.3, -2.2, -1.1, 0, 1.1, 2.2, 3.3$ V, $V_{nw} = 0$ V

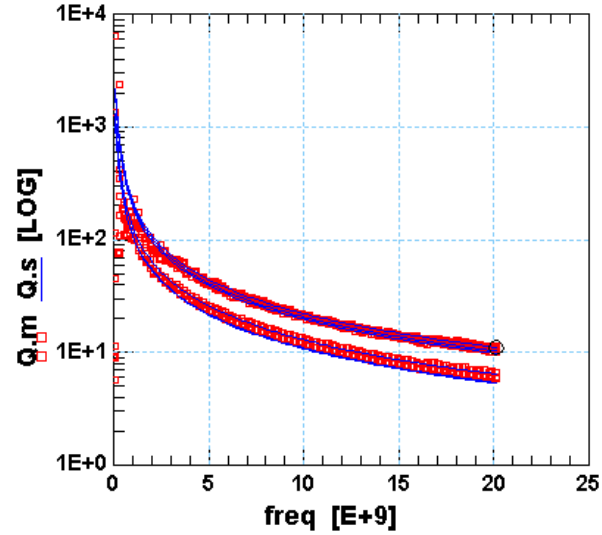


Figure 2-7 VAR_301:Q vs. Freq at $V_g = -3.3, -2.2, -1.1, 0, 1.1, 2.2, 3.3$ V, $V_{nw} = 0$ V

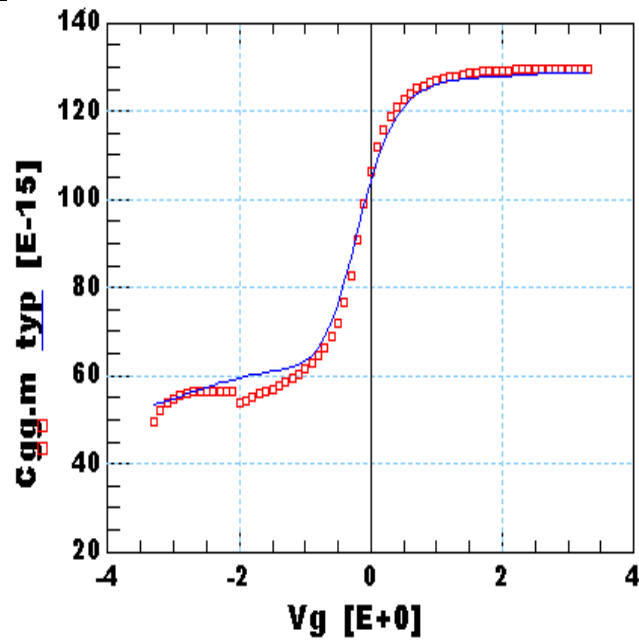


Figure 2-8 VAR_301: C_g vs. V_{gb} at Freq=5.9GHz

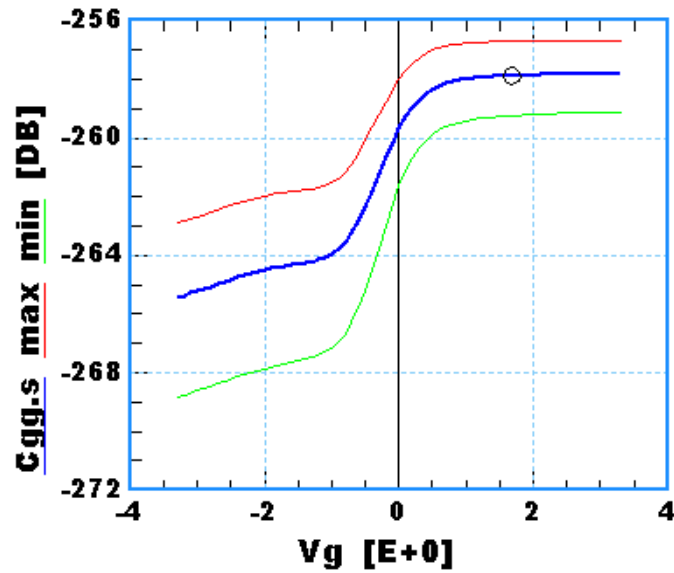


Figure 2-9 VAR_301: C_g vs. V_{gb} at Freq=5.9GHz for center model

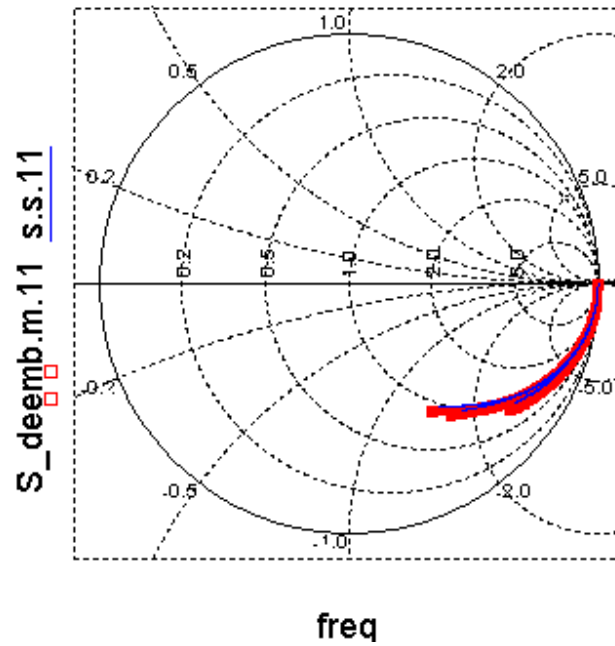


Figure 2-10 VAR_301:S11 at $V_g = -3.3V$ to $3.3V$, $V_{nw} = 0V$, Freq=100MHz to 20.1GHz

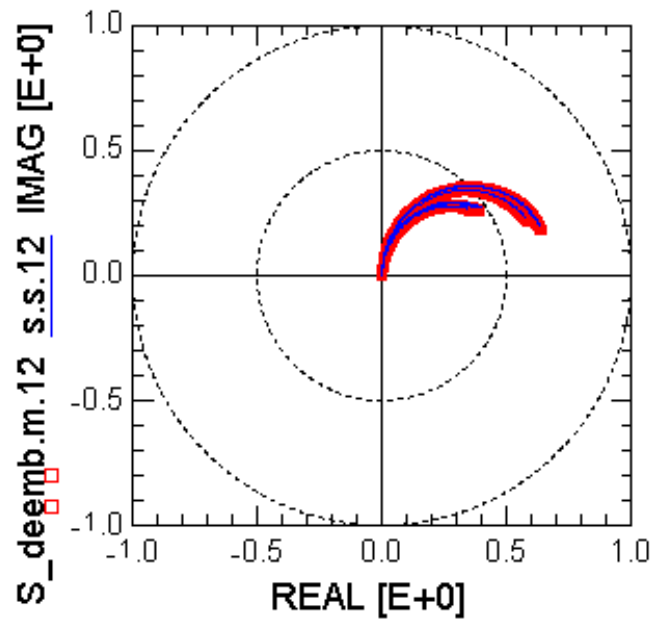


Figure 2-11 VAR_301:S12 at $V_g = -3.3V$ to $3.3V$, $V_{nw} = 0V$, Freq=100MHz to 20.1GHz

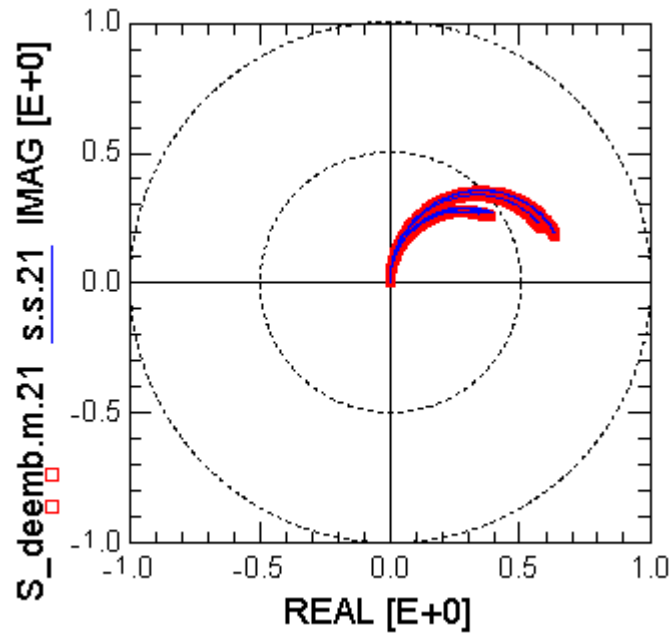


Figure 2-12 VAR_301:S21 at $V_g = -3.3V$ to $3.3V$, $V_{nw} = 0V$, Freq=100MHz to 20.1GHz

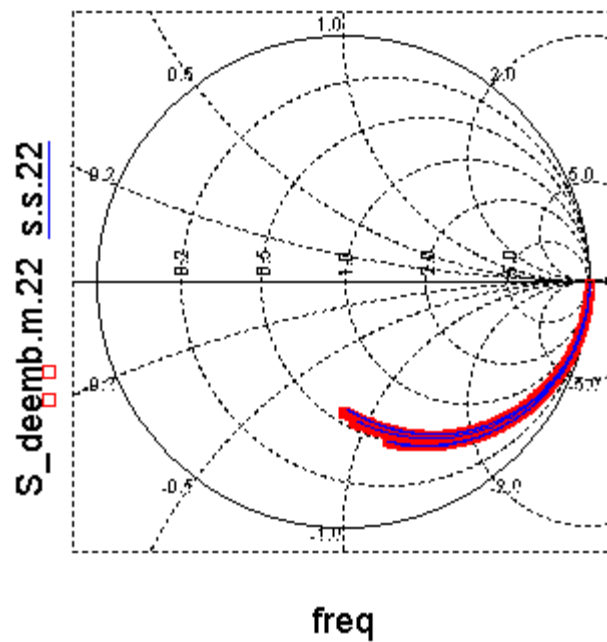


Figure 2-13 VAR_301:S22 at $V_g = -3.3V$ to $3.3V$, $V_{nw} = 0V$, Freq=100MHz to 20.2GHz

VAR_304

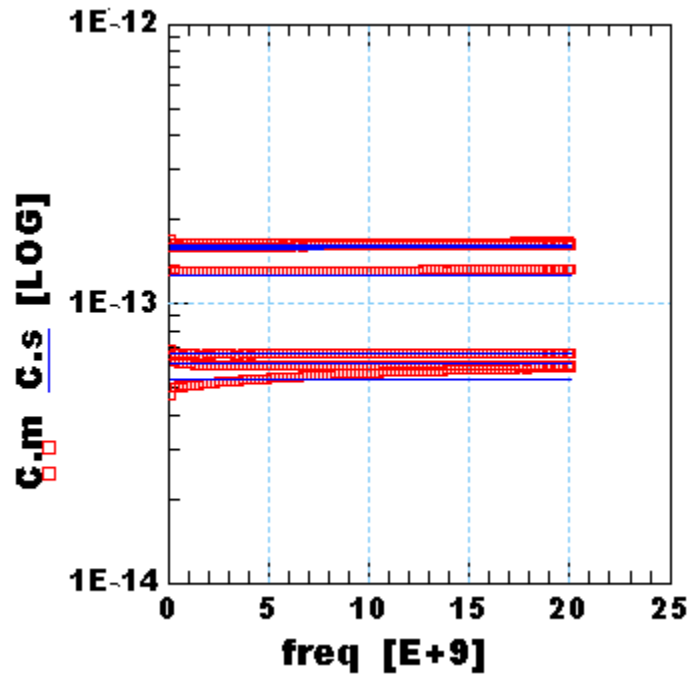


Figure 2-14 VAR_304:Cg vs. Freq at Vg=-3.3, -2.2, -1.1, 0, 1.1, 2.2, 3.3 V, Vnw=0 V

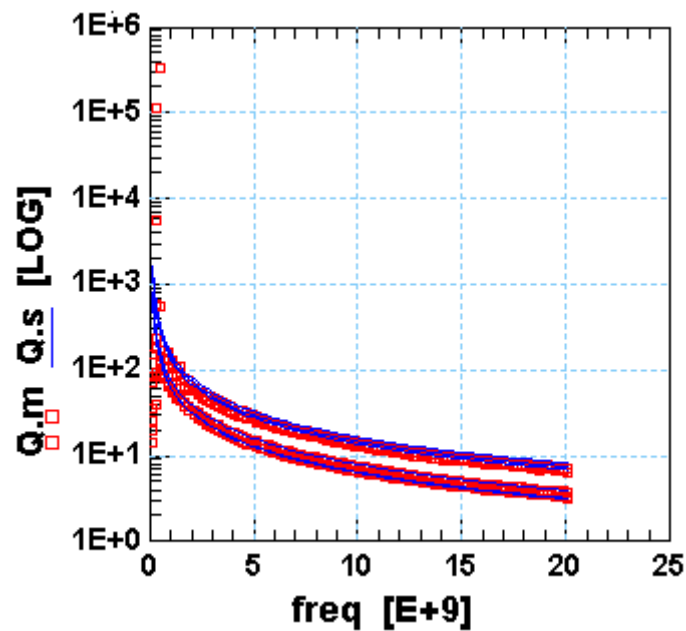


Figure 2-15 VAR_304:Q vs. Freq at Vg=-3.3, -2.2, -1.1, 0, 1.1, 2.2, 3.3 V, Vnw=0 V

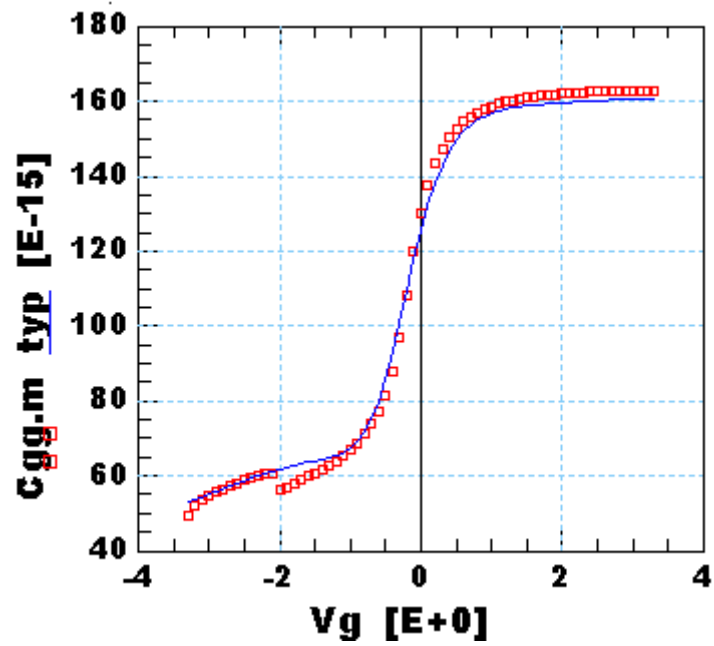


Figure 2-16 VAR_304:Cg vs. Vgb at Freq=5.9GHz

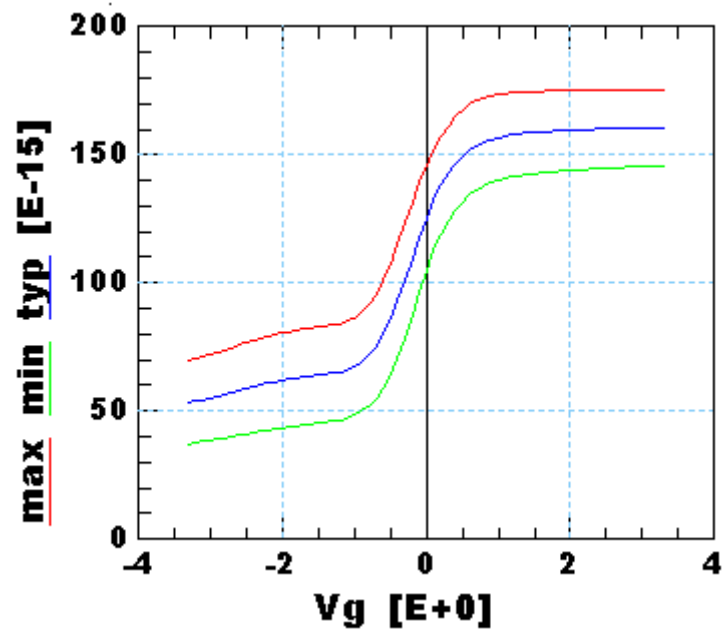


Figure 2-17 VAR_304:Cg vs. Vgb at Freq=5.9GHz

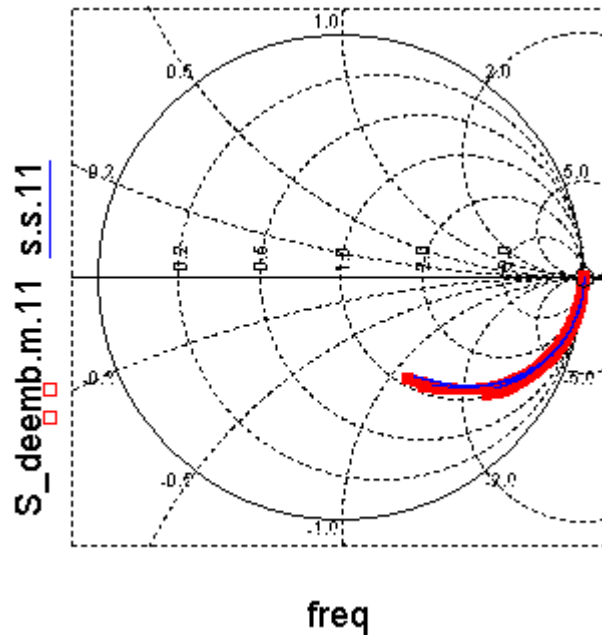


Figure 2-18 VAR_304:S11 at $V_g = -3.3V$ to $3.3V$, $V_{nw} = 0V$, Freq=100MHz to 20.1GHz

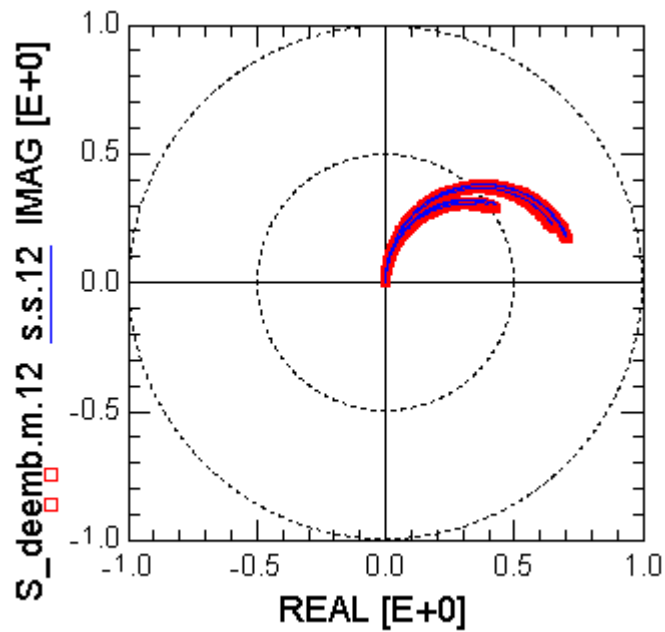


Figure 2-19 VAR_304:S12 at $V_g = -3.3V$ to $3.3V$, $V_{nw} = 0V$, Freq=100MHz to 20.1GHz

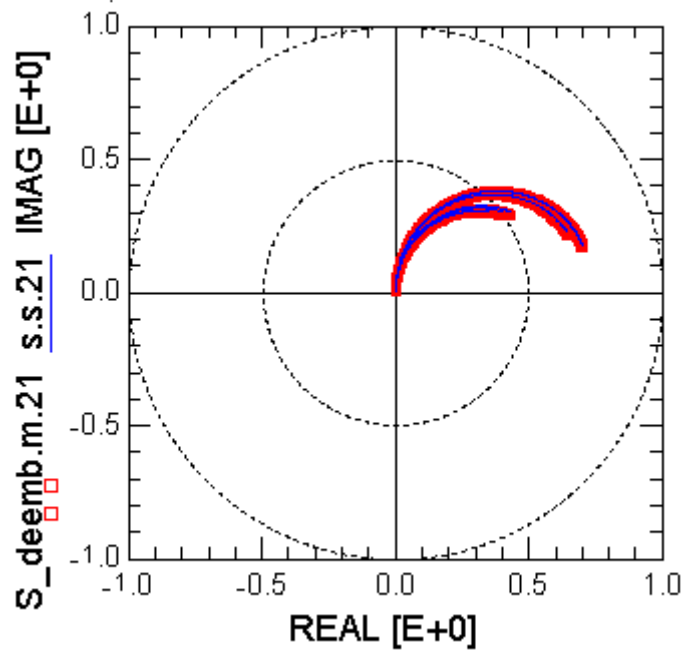


Figure 2-20 VAR_304:S21at $V_g = -3.3V$ to $3.3V$, $V_{nw} = 0V$, Freq=100MHz to 20.1GHz

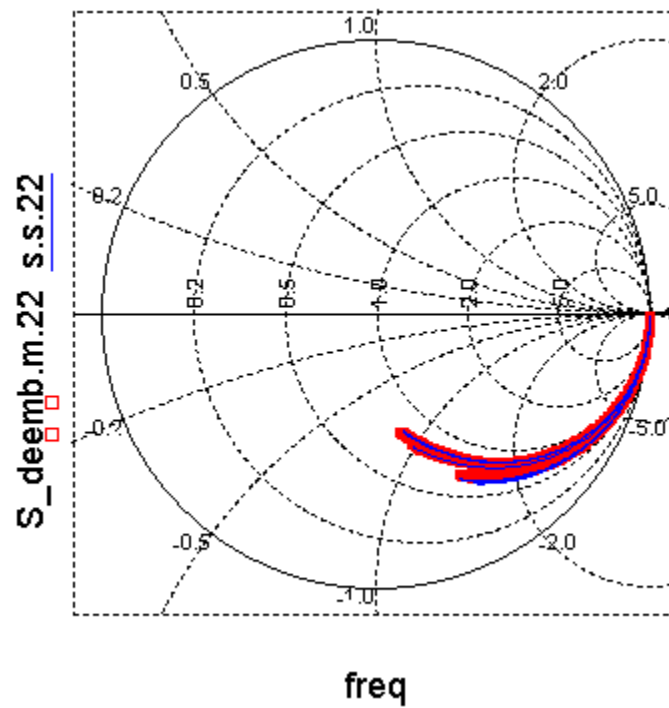


Figure 2-21 VAR_304:S22 at $V_g = -3.3V$ to $3.3V$, $V_{nw} = 0V$, Freq=100MHz to 20.1GHz

VAR_305

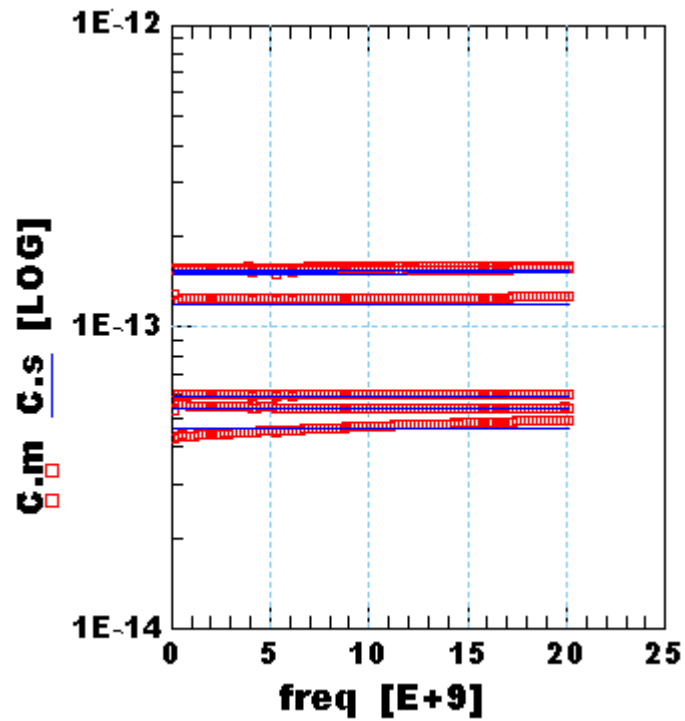


Figure 2-22 VAR_305:Cg vs. Freq at $V_g = -3.3, -2.2, -1.1, 0, 1.1, 2.2, 3.3$ V, $V_{nw} = 0$ V

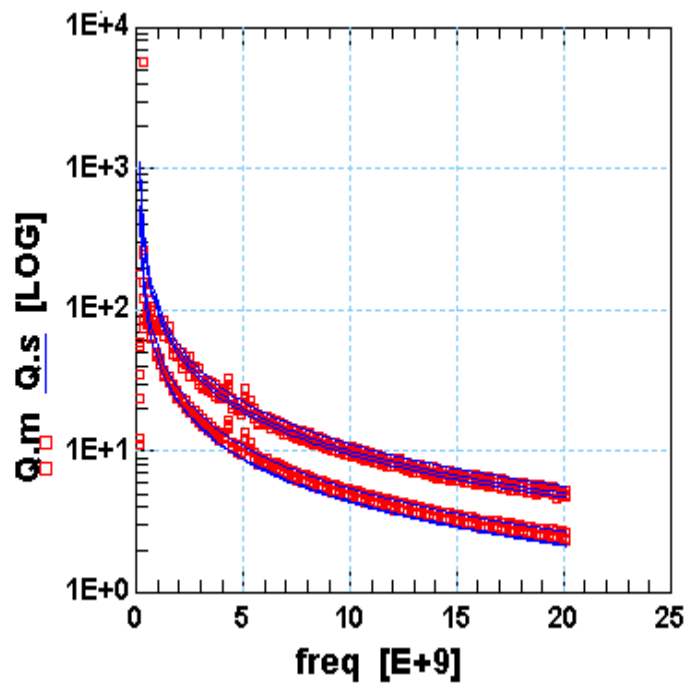


Figure 2-23 VAR_305:Q vs. Freq at $V_g = -3.3, -2.2, -1.1, 0, 1.1, 2.2, 3.3$ V, $V_{nw} = 0$ V

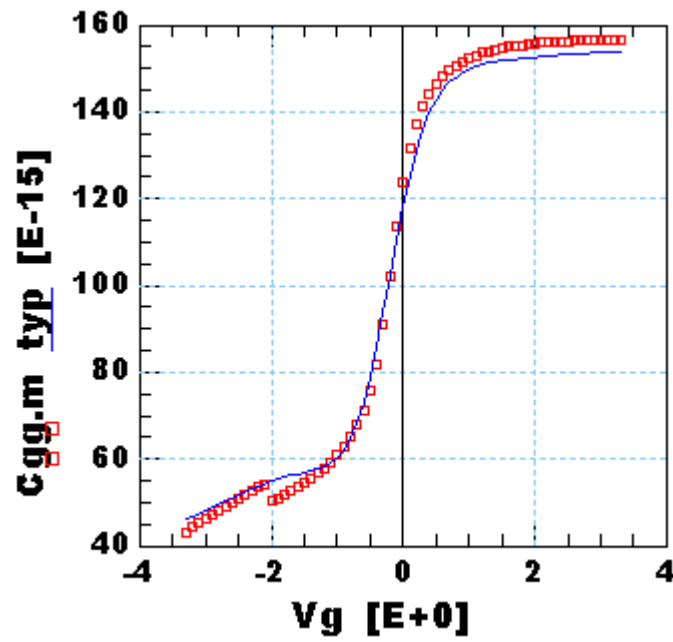


Figure 2-24 VAR_305:Cg vs. Vgb at Freq=5.9 GHz

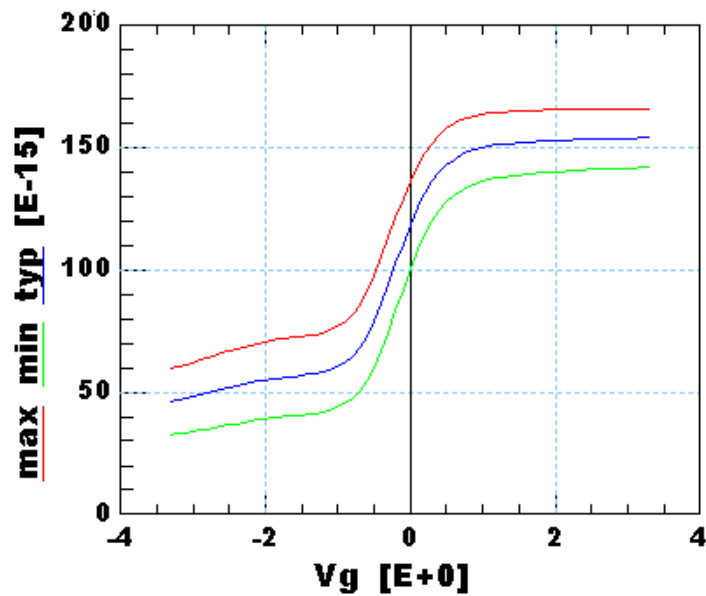


Figure 2-25 VAR_305:Cg vs. Vgb at Freq=5.9GHz

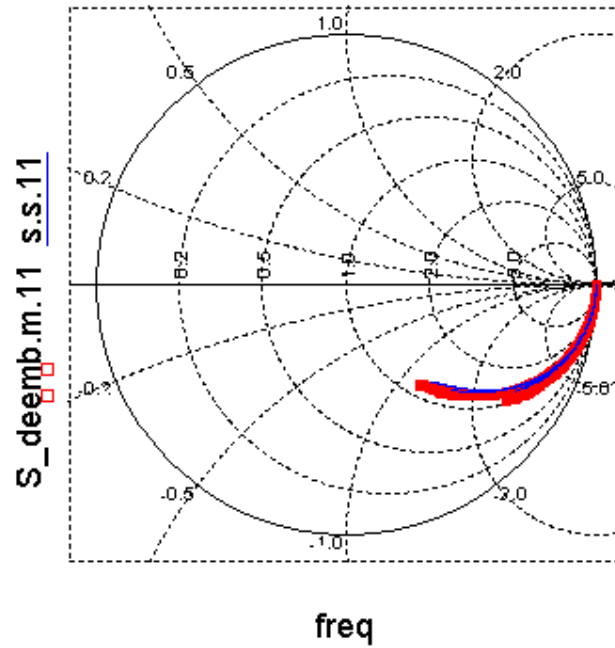


Figure 2-26 VAR_305:S11at $V_g = -3.3V$ to $3.3V$, $V_{nw} = 0V$, Freq=100MHz to 20.1GHz

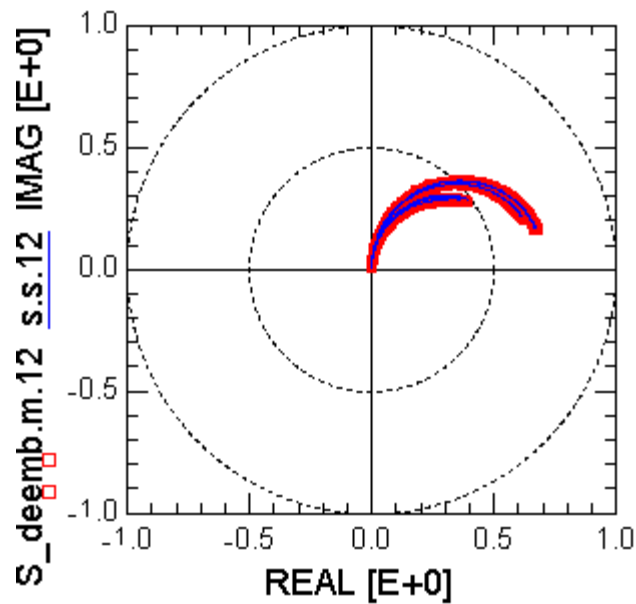


Figure 2-27 VAR_305:S12 at $V_g = -3.3V$ to $3.3V$, $V_{nw} = 0V$, Freq=100MHz to 20.1GHz

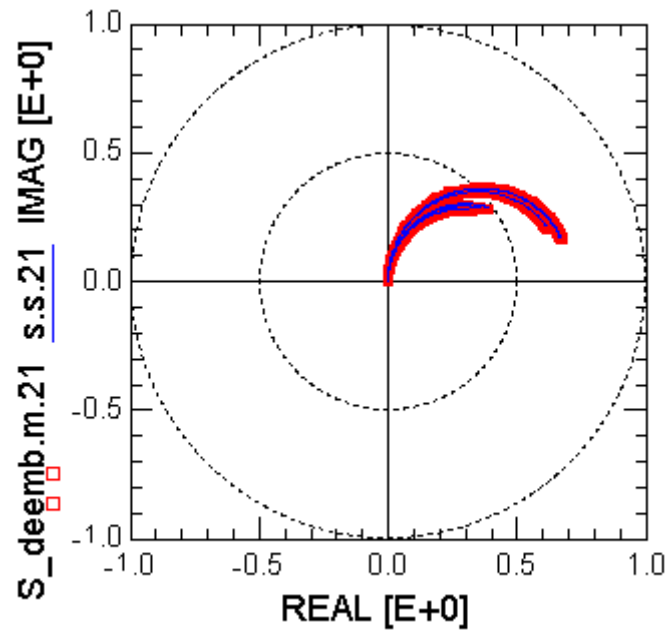


Figure 2-28 VAR_305:S21 at $V_g = -3.3\text{V}$ to 3.3V , $V_{nw} = 0\text{V}$, Freq=100MHz to 20.1GHz

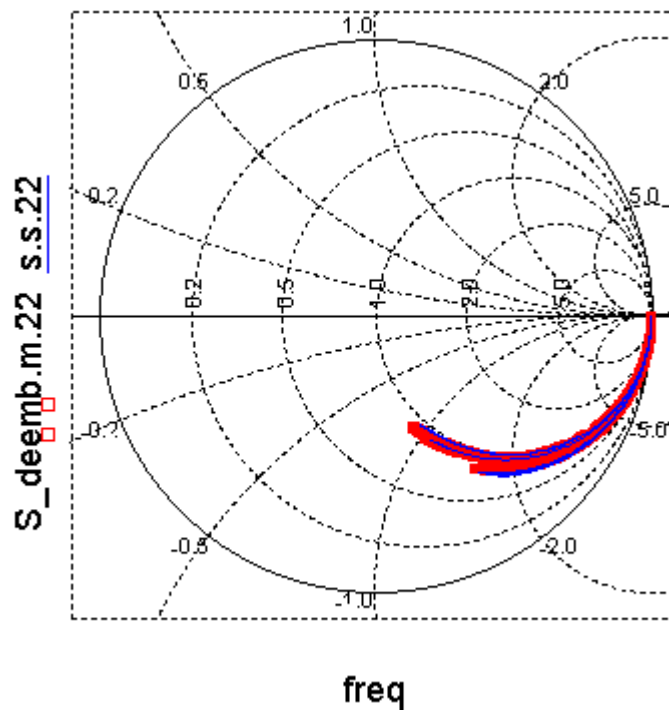


Figure 2-29 VAR_305:S22 at $V_g = -3.3\text{V}$ to 3.3V , $V_{nw} = 0\text{V}$, Freq=100MHz to 20.1GHz

3 Appendix

3.1 Hspice Warning Message

Table 3-1 Hspice warning message

No.	Warning Message	Explanation
1.	None	None

3.2 Eldo Warning Message

Table 3-2 Eldo warning message

No.	Warning Message	Explanation
1.	None	None

3.3 Spectre Warning Message

Table 3-3 Spectre warning message

No.	Warning Message	Explanation
1	None	None